

Institute of Actuaries of India

Subject SA4 – Pensions and Other Benefits

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) What are the risks of this scheme the subscribers should keep in mind?**

- The defined contribution nature of the scheme means the ultimate outcome of benefit amount is unknown. [0.25]
- The scheme is only focussed on retirement at age 65. It is not flexible to changing needs of the citizens. [0.5]
- The death benefits are only a return of contributions. No interest on that is mentioned, therefore a potential loss to the subscriber's beneficiary [0.5]
- The mandatory annuitisation portion at retirement with an insurance company means there is a risk that the value of the investments do not move in line with annuity rates. [0.25]
- The lifecycle investment approach comes with a risk that the asset allocation percentages set by the design may not be optimal in terms of performance, especially with respect to risk of timing of the asset allocation switching that will happen [0.5]
- The subscriber is at risk of choosing in appropriate asset allocation or fund manager that could affect their ultimate outcome [0.25]
- Investment charges may change in the future [0.25]
- Investment charges are not disclosed and so the subscriber is at risk of lower returns due to higher expenses, which they would not know about [0.5]
- Tax rates and the tax allowance may change adversely in the future due to future government decisions [0.25]
- The contributions are fixed for the whole membership and one would be at risk of the inflationary erosion of the value of these contributions not increasing with age. [0.25]

[3]**ii) What are the risks for the regulator/government?**

- Citizens may not understand the scheme nuances very well which will be a barrier to participation [0.25]
- The distribution channel (and how they are remunerated) of how this plan will reach citizens will be critical. Risk of mis-selling need to be mitigated through empanelment / trainings, regulations and governance. [0.5]
- There is a risk that many subscribers do not receive benefit outcomes as they might have thought. This is a high reputational risk. E.g poor investment performance or higher charges [0.5]
- There is a risk that the choice of operating model / administration is not done properly resulting in incorrect values being maintained and given out to members. [0.5]
- There is a risk that the administration charges are not as efficient as possible [0.25]
- If it is envisaged that the scheme will be managed by empanelled partners (e.g. for administration, investment) there is a risk whether suitable players are willing to participate in the eco system. The remuneration structure will be critical to balance between efficient cost whilst having suitable players interested. [0.5]

[2]**iii) Suggest areas where the design and features of the plan could be evolved to boost participation. In particular keeping a philosophy of flexibility and choice in mind.**

For each suggestion state key pros and cons for the subscriber and government.

Eligibility

- One could allow eligibility up to age 65 [0.25]

Pros/Cons

- This is good for subscribers who want to start late [0.25]
- This may not be suitable unless the retirement age is deferred in such cases, else there may not be enough time for a meaningful membership period (accumulated sum), especially for new subscribers [0.5]
- It exposes the risk of adverse investment performance in a short period of time that can't be made good later as it is too close to retirement [0.25]

Contribution Rates

- One could allow members to change their contribution rates a certain number of times in their membership [0.25]
- Or have increasing rate of contribution by age [0.25]
- Have contribution rates indexed with CPI each year or a % of earnings [0.25]
- Allow reduction of contributions as well to enhance flexibility [0.25]

Pros/Cons

- The concept of any increasing contribution ensures a maintenance of some value with inflation as people get older [0.25]
- However, people may not be able to afford automatically increasing rates (by age or if indexed) [0.25]
- Members will benefit from flexibility in choosing contribution levels upwards and/or downwards to cater to other needs, if required [0.25]
- For the Government all this would mean considerable additional administration burden and costs to the system to develop/maintain IT platforms etc.. [0.5]
- Some Members are also less likely to understand the rationale for changing rates and the increased choice, thereby getting confused [0.25]

Exit

- Flexible retirement age should be introduced. For example any age post 60 and before 70 [0.5]

Pros/Cons

- Added administration for the system. It would need to be decided whether the choice is exercised at the start or nearer retirement. [0.25]
- Greater flexibility for members to choose per their personal circumstances [0.25]
- Government will want to ensure that the earliest age is not too low so as not to compromise with the main purpose of the scheme of a retirement plan. [0.25]
- One could add a minimum membership term before becoming eligible for pre age 65 retirement [0.25]

- Flexibility to defer retirement [0.5]

Pros/Cons

- This helps to mitigate risk of poor investment performance near the original retirement age [0.25]
- Suitable communications would need to be done to members [0.25]
- Further administration burden [0.25]
- Members may not make an appropriate choice, in particular not deferring when they should if poor investment performance or deferring and then realising they require the income sooner. [0.25]

Benefit

At retirement a choice of the form of pension as well as the insurance company should be given. This is not specified in the scheme as of now. E.g. Joint life pensions / Guarantee periods [0.5]

Pros/Cons

- Government will need to empanel insurance companies [0.25]
- Member education will be required [0.25]
- Government will need to have greater governance in terms of ensuring value for members annuities e.g. can they negotiate some incentivised rates [0.25]

Provide beneficial tax treatment for annuity purchases or annuity income e.g. no tax on annuity purchases or reduced tax rates for annuity income [0.5]

Pros/Cons

- Better value for the members [0.25]
- Government would need to convince and incentivise insurers to provide this [0.25]
- Any tax concessions are a cost to the Government, albeit many years into the future [0.25]

Provide flexibility on the term over which the lump sum is received and/or time period where the payment is deferred (e.g. up to three years) [0.5]

- Same goes for the purchase of the annuity

Pros/Cons

- Better flexibility for members to personal circumstances, as well as mitigate timing of investment markets [0.25]
- Added admin burden for the system [0.25]

Provide access to accumulated funds in case of personal exigencies during membership [0.25]

Pros/Cons

- Will be appreciated by members [0.25]
- Tight controls and criteria would need to be laid down so the main purpose of the plan is not diluted e.g. limitation of amounts to be accessed and circumstances when permitted [0.25]

- Tight governance on administration procedures and preventing moral hazard need to be implemented [0.25]

Introduce an income drawdown approach to accessing the funds post retirement [0.5]

Pros/Cons

- Will be appreciated by members for flexibility [0.25]
- Ensures members can still benefit from investment performance in retirement [0.25]
- Strong advisory requirement would need to be put into place to prevent incorrect amounts being drawn down and funds being depleted too quickly or not appropriately. [0.25]
- Increased administrative burden [0.25]

Investments

- Allow members to choose their asset allocations themselves

Pros/Cons

- Greater flexibility for members
- Greater member education would be required
- A declaration of the member acknowledging they have understood the risks and/or have taken suitable advice may be required to mitigate risks for the Government down the line

The asset buckets can be increased e.g. have a number of “sector” indices to choose from or in the case of bonds introduce corporate bonds and different term of bonds

Pros/Cons

- The more choice the more complex it becomes for members to understand
- A declaration of the member acknowledging they have understood the risks and/or have taken suitable advice may be required to mitigate risks for the Government down the line

Introduce a minimum assured return

Pros/Cons

- Would be very popular to members
- Design of a minimum return would be complex and fund managers may not agree given additional capital requirements
- Government would need to draft complex regulations around solvency / capital measures to ensure guarantees are valued and provisioned appropriately.
- Government would need to decide if it is stipulating the guarantee or letting fund managers design it
- Any guarantee would need to be stipulated from a year on year basis or overall accumulation period method.
- Any guarantee would need to be at a level that is meaningful to members but balancing the additional cost

[15]

iv)

The Government wants to introduce a special scheme for corporates to plug and play into the national scheme.

State key areas the government will need to consider when thinking about introducing this

0.75 mark each area

For each area, suggest two specific options that might be implemented.

0.25 per each suggestion

Eligibility

- What type of companies will be able to participate?

For example

- There may be a different model for the public government owned companies and the private companies

Suggestions

- It may mandate participation for new entrants from a certain date for government owned companies
- Keep it voluntary for private sector companies
- Introduce auto enrolment with an opt out for corporate employees

Contribution rates

- Will the Government require contributions from the employer?

Suggestions

- Private Companies are required to co-contribute a minimum of 50% of the contribution on the employee's behalf (i.e. contribution matching) up to a maximum of 500 per month employer contribution
- For Government owned companies this may be 100% by employer

Taxation treatment Employees

- How will tax allowances be treated if the employee is participating through a corporate

Suggestions

- Any employer contribution should not be treated as income for the employee with any cap
- If an employee participates through the corporate then his/her contribution should also be tax deductible up to a limit at least

Taxation treatment Employers

- How will tax allowances be treated if the employer is participating

Suggestions

- Any employer contribution should be tax deductible as business expenses for the employer
- Any costs incurred in facilitating the scheme for employees can also be treated as business expenses

Corporate Enrolment

The Government will need to set out the procedure for corporates to facilitate enrollment of its employees into the scheme

Suggestions

- Registration of the corporate so it can facilitate contributions to the scheme
- Issuance of a Corporate Number so member's accounts can be tagged to

the particular corporate

Corporate Decisions/Choices

- Will the Corporate be required (or have an option) to enforce certain choices for its employees i.e. make its own plan features from the various choices within the existing scheme architecture

Suggestions

- Corporates may choose the level of contributions applicable to its employees or different contribution levels for different set of employees
- Corporate may choose the asset and fund management choices for its employees or at least limit the choices as a default choice

Treatment of Members if they leave a corporate

- How will member accounts be treated on leaving service

Suggestions

- Accounts are maintained as continuous within the scheme. Members can continue to contribute on their own or through their new employer
- A new member account is opened and the previous account is transferred into the new one

Governance around Employee Communications

- The Government will need to decide if some standards and controls to be put in place to ensure employees receive consistent and accurate information about the scheme via the Corporates

Suggestions

- Lay down standardized communication material to be made available via the Corporate about the scheme, including promotional material
- Require a Corporate to appoint an external qualified and registered retirement expert to educate employees about the scheme

[10]

v)

The proposal is that each employer must pre-fund the benefit by paying 4% of monthly salary to a centralised government administrator. Employers will still be ultimately liable for the benefit obligation / payment and would receive reimbursement from their account held with the central office. Employers may apply for an exemption and run their own separate Trust.

State the key potential implications, considerations and questions for employers of the above proposal.

Overall principle / risk

- The proposal will bring a hybrid approach to meeting the cost of the benefits with regular defined contributions but with an ultimate defined benefit risk still with the company. [0.5]
- This will help reduce volatility of cash requirements to fund benefit payments [0.5]

- in the longer term as country is going from a pay as you go to pre funding
- However, for companies that are newer, or in industries with a high level of attrition, it means having a cashflow outlay for a benefit that would otherwise not be payable for a long time. This reduces the company's ability to optimise its cashflow, if needed. [1]
- The company will need to ultimately assess its exposure to whether a 4% contribution plus any investment earnings credited by the Government will be sufficient to meet the benefit payments. This introduces the concept of having to conduct actuarial assessments. [1]

To apply for exemption or not

- One key decision will be whether the company applies to have its own Trust or pay to the Central Administrator [0.5]
- This will depend on some key guidance/regulations as follows:
 - The time and cost to set up a Trust [0.25]
 - Eligibility criteria to make an application e.g. profitability of the company, any concessions for the size of the company [0.5]
 - What stipulations will the company have over deciding administration and investments of the Trust? [0.25]
 - What investment restrictions may there be? [0.25]
 - What are the Trustee requirements; both for eligibility to be Trustee and also roles and responsibilities [0.5]
 - Are there employee communications / disclosure requirements about the Trust that need to be made available [0.25]
 - What are the financial and other reporting requirements of the Trust to any Regulator? If these are far more than under a Trust then this is a deterrent. [0.5]
 - Will the exemption require companies to promise benefits at least of the same value or have a minimum contribution rate at all times. This is a key point because if there is a minimum contribution for exemption then the opportunity to leverage the actuarial assessment and compare it against the 4% minimum will be greatly reduced and be a point against trying to go for an exemption. [0.5]

The above points will play a key role in weighing up whether it is worth managing the Trust, which will not be a core business to them and will take away time from their core business. [0.25]

- Each corporate will need to conduct a feasibility after taking the following account:
 - 4% of Salary contribution vs an assessed longer term cost of the benefit in an actuarial assessment. [0.25]
 - An actuarial assessment may be required anyway if the country has accounting standards that work on the accrual principle. Otherwise the company could do one to assess if the contributions would likely be less than 4%, given their demographic profile. [0.5]
 - One will need to establish if the past service needs to be funded and how that calculation compares to the 4% of salary. [0.5]

- Compare administration fees to run a Trust compared to anything the Government may charge, if any. [0.25]
- This includes cost of appointing a third party administrator if required [0.25]
- Compare the tax treatment of the contributions into the Trust versus the Central fund, and also any investment earnings from either vehicle. The decision may be simpler not to set up a Trust if there is only a tax allowance of up to the 4% of Salary and pay as you go payments above that limit, instead of full pre funding. [0.25]
- The Trust may be able to obtain investment returns better than the Government scheme which, all other things being equal, will bring the ultimate cost down for the Company. [0.5]

The company will need to keep in mind any historical record of the Government's ability to implement the administration and servicing of other social security schemes / compliance to gauge if the employer is likely to face ongoing day to day issues dealing with the Central administrator in getting money/ information etc.. on time to meet the company obligations – as it is still going to be responsible to pay the benefits. [0.5]

[10]

vi)

Under the exemption reporting rules for all companies, the Government has stated that an actuarial valuation at least every three years must be submitted to the centralised regulator to demonstrate that the Trust is providing sufficiently for the benefits stipulated.

Comment on the considerations and options the regulator will have in setting the regulations for such a valuation

Government must decide the purpose of the Valuation

- A minimum must be to demonstrate the funding status [0.25]
- It may also include the contribution requirements, compared to the statutory requirement [0.25]

Valuation principle

- Will the valuation be on a going concern principle or a discontinuance basis or both? [0.5]
- The approach and all aspects of the valuation should be consistency between companies and enable comparability. [0.5]
- This will not always be appropriate given different industry segments and economic scenarios and so a balance will need to be factored in setting prescriptive requirements [0.5]
- The level of prudence desired will need to be decided when setting the guidelines [0.5]

Valuation Method

- Given the scheme is salary related, mandatory for all companies over 10 employees, it would make sense to have the Projected Unit Credit method as a base valuation method. [0.5]
- This ensures that salary is projected to the time of exit and will reflect year on year accrual closer than prospective funding methods. [0.5]
- However, when considering the fact there is a statutory contribution rate and one would need to establish whether there is a surplus/deficit when using that rate – it may be suitable to use an Entry Age method. [0.5]
- Entry Age will give a higher liability value if salary escalation is lower than the discount rate – to give higher security. [0.5]

Valuation Assumptions

- The following key assumptions will be required: [0.25]

Discount rate

- A risk free rate with a stipulated source will aid consistency and comparability. [0.25]
- A risk free rate will also mean a higher level of security by virtue to higher calculation of liabilities (if actual assets are in more riskier assets and expected to earn more) [0.5]
- The Discount rate can be chosen to reflect a risk premium on the assets held. [0.25]
- The Rules will need to apply to exempt Trusts as well as companies where the funds are with the central administrator [0.25]
- A risk premium range for each asset class can be published by the regulator periodically. This will keep some level of consistency rather than have each company develop their own rate freely [0.5]
- Need to decide if it will be a market related rate or long term rate. Market rate approach can give a more realistic reporting but comes with the risk that it is subject to volatility. [0.5]

Salary Growth and Attrition rates

- One stipulated rate may not be appropriate given different industries and economical scenarios. [0.25]
- One can give a relation to inflation guidance as an option [0.25]
- One could publish ranges from all the valuations the Regulator receives and collect data from all companies on a periodic basis. [0.5]
- Any deviations from ranges should require the company to provide a comprehensive explanation and data to back up their assumptions [0.25]
- The guidance will need to reflect that companies may apply different salary growth for different category of staff. Guidance around that will need to be provided e.g. Liability weighted ranges and reporting [0.5]
- For guidance it can be simply a minimum floor guidance that is given. [0.25]

Mortality

- From time to time a standard mortality table should be published for all companies to use [0.25]
- Given the scheme is a lump sum plan this is not a material assumption. [0.5]

Asset Valuation

- Need to consistent with the liability calculation approach [0.5]
- The regulator will need to be prepared to value the assets on this basis for all companies who hold funds with the central administrator as an input to the valuation exercise. [0.25]

Funding Status

- The Regulator will need to decide if there will be a minimum funding level to be maintained [0.25]
- It will need to provide details of what time period a company has to make up any shortfall compared to the minimum level. [0.5]
- One needs to decide what happens for companies that are paying the statutory 4% contribution in the case of deficits versus those with a Trust. [0.5]

Valuation Data

- Stipulate the data checks and certification needed as part of the exercise [0.25]
- Provide a standardised template for data collection [0.25]

Reporting

- Outline the format of the valuation report, that is consistent with actuarial professional guidance [0.25]
- Stipulate what “certification” and “Returns” are needed to be filed with the Regulator, which can be done online for instance. [0.25]

[10]**[50 Marks]****Solution 2:**

i) Possible sources of deficit [For each valid point 0.25 marks]

a) The actual experience is worse than the assumptions used in funding plan

- Longevity of pensioners higher than assumed
- Death (in service) experience of members higher than assumed at higher ages, say after 45
- Withdrawal /early retirement experience lower than assumed
- Death in service experience lower than assumed at lower ages, say below 45
- Investment return lower than the assumed returns
- Inflation of salaries higher than assumptions
- Changes in the profile of the group e.g. changes in the average age due to new entrants being different profile than leavers.,
Changes in the retirement age that was not considered in the valuations
- Changes in the marital/dependent profile/age difference
- Changes in the take up option for example transfer value, early retirement, commutation [2]

b) Changes in the parameter values used in valuation- Strengthening of basis

[For each valid point 0.25 marks]

- lowering discount rate

- using lighter mortality table for pensioners/actives

- enhancing salary escalation assumption

- reducing withdrawal/early retirement rate [1]

c) Benefits: paid higher than the benefits valued in funded -due to Changes in the benefits (e.g. changes in the minimum pension, accrual rate, salary definition, liberal commutation terms, and liberal discretionary pension increases etc.) [0.5]

d) Contributions paid lower than recommended rate – contribution Holiday [0.5]

e) Other reasons: [0.25 for each valid point]

- Bulk transfers changing the funding level
- Bulk transfers changing the membership mix
- Changes in the funding method
- Changes in Tax laws (e.g. introduction of GST) and therefore a lowering of investment returns
- Existing deficit growing (Interest on existing deficit)

[5]

ii) Issues to be discussed during the meeting with the Trustees

The objective of the meeting is to

- Understand the causes of growing deficit [0.25]
- Verify that the objective of the Trustees to administer the scheme is consistent with the funding objectives of the company [0.5]
- Understand the impact of company initiated activities on funding plan (e.g. mergers /de mergers/, introducing VRS, lateral entries permitted in the pension Scheme) [0.5]
- Understand the impact of external changes (tax laws, external changes etc.) [0.25]

Assets:

The composition of asset portfolio and the split between the various asset classes & the actual performance of the each asset class [0.25]

How the assets are managed? -If assets are managed by external fund managers, the mandate given to the fund manager -If assets are passively managed, examine the bench mark/index used to track the investment performance [0.5]

Method of valuation: Whether assets are taken at book value or market value. If any other method is used– how the changes in market value are reflected in valuation (e.g. if book value approach is used, check the provision for amortization) [0.5]

Examine the Provision given NPA & investment related expenses & ensure that they are not excessive. [0.25]

Quality of Assets: Discuss whether there any investments that are not “rated” & examine performance of these assets & the volatility associated with it -Examine the investment performance of the assets relating company X held by Trustees(self-investment) [0.5]

Liabilities

The composition of the liabilities split between actives/deferred/pensioners	[0.25]
& how the composition has changed over the period	[0.25]
Any investigation carried out by the Trustees to understand the reasons for the change, if any	[0.25]

Benefits

Are the benefits valued in line with the benefits stated in scheme rules by company X	[0.25]
What is the Trustees' approach for discretionary benefits, or minimum pensions, if any?	[0.25]
Are these benefits are valued for funding purpose or being paid as "PAYG" approach	[0.25]
What is the power of Trustees in changing the form of benefits (e.g. commutation factors, benefits payable on early or late retirement?), whether or not they may be actuarially neutral	[0.5]
Where the Trustees have power in effecting changes in such benefits, whether such decision, if any is shared with Company X	[0.25]

Retirement age:

It seems the retirement age has been reduced during 2017-18	[0.25]
Whether any investigation carried out to understand the impact of lowering retirement age and whether additional financing was called for from company x?	[0.5]

Profile of members:

The average age of members increasing	[0.25]
How does the age profile of each category of members vary over the period?	[0.25]
Are there any special category of members joining (eg lateral entry, Top Executives joining at higher age)-thereby altering the profile of members	[0.5]
Was the profile of members get altered due to the merger/acquisition	[0.25]
Whether the salary profile of members is consistent with salary definition used for pensions?	[0.25]
Was there any investigation made to understand the changing age/salary profile of members to understand its impact on funding?	[0.5]

Data used

Company X will be providing the members data to value the liabilities which needs to be validated with the data provided by Trustees to previous actuary.	[0.5]
What is the mechanism with Trustees to get and validate the data to ensure its relevance, completeness & accuracy?	[0.5]
Whether mechanism is in place to reconcile the movements & to ensure consistency.	[0.25]

What is the approach in allowing for errors in data? Is any explicit provision kept? [0.5]

Parameters

How the basis for various parameters such as mortality/longevity/resignation, early retirement have been developed? [0.5]

If industry data/mortality tables are used to set the parameters, how they are modified to reflect the changing profile of members? [0.5]

Are salary growth assumptions reflecting the salary profile of members? [0.25]

What is the Trustees' approach in choosing the discount rate? Market consistent basis or Asset based approach? [0.5]

The level of margin kept for the random variations of the scheme experience [0.25]

Frequency of review of the various parameters and the power of Trustees to set the parameters (in consultation with the Scheme Actuary) [0.5]

Method of valuation

It is not clear whether valuation by scheme actuary has been done for accounting purposes or for funding purpose. [0.5]

If valuation is done for accounting purpose, then actuary might have used PU method [0.25]

If any other method is used in valuation, rationale for using the same and its financial impact (on deficit) needs to be studied [0.5]

One should discuss the Company/Trustee's previous funding strategy and any philosophy they have agreed to follow (e.g. maintain 100% funded on a certain basis/time period or have they decided to reduce contributions or they are not funding for making up any past deficit). [0.25]

Provisions

Any specific provisions are kept in the valuation of liabilities? For example, provision for unhedged risk, expenses, data deficiency etc. [0.5]

How significant these provisions, if any are as compared to the total liabilities? [0.25]

Contributions

What has been the recommended contribution for various years in the past as per Scheme Actuary & what is the actual contribution paid by company X during these years? [0.5]

Other issues:

Any other factors contributing to growing deficit? [0.25]

[12]

Documents

Copies of the following:

- Scheme Rules together with Amendments, if any [0.25]

- Trust deed [0.25]
 - Valuation Reports of last 3 years [0.25]
 - Data used in the valuation of last two years [0.25]
 - Reports relating to any investigations carried out [0.5]
 - Last 4 years' Trust Financial statements [0.25]
 - Asset Schedule [0.25]
- [2]**

iii) Professional guidelines to be followed by a Consultant Actuary while taking the new assignment

1. Professional Conduct Standards (PCS) sets the standards to be followed when taking new assignments and on commenting the role of scheme actuary [0.5]
2. The Company has the indisputable right to take second opinion and/or to have separate advisors for different matters [0.5]
3. The consultant actuary should contact the Scheme Actuary and inform him of the assignment he proposes to undertake seeking his opinion whether there are any professional reasons not to accept the appointment or any particular considerations which ought to be borne in mind before giving advice [0.5]
4. The Scheme Actuary, upon receiving such an enquiry, must provide the required information as soon as possible and should make no attempt to obstruct the company who wishes to appoint advisors [0.5]
5. Both the actuaries must avoid any action that would unfairly injure the professional reputation [0.5]
6. Criticism of one member's work by another member is acceptable, provided that the criticism is properly reasoned and believed to be justified [0.5]
7. Where criticism of an actuary's work is made, care must be taken to acknowledge that the other actuaries may quite properly hold different professional opinions and that special circumstances may exist in any particular case [0.5]

[3]

iv) Factors to be considered while setting the discount rates:

The discount rate represents the rate of interest with which the benefit, expenses and contribution cash flows relating to future period are discounted to place a value on liabilities & to derive the contribution requirements. [0.5]

While choosing the discount rate, the basic principle will be that the value placed on benefits needs to be consistent with value of assets. In case value of assets is taken as book value, then asset based discount rate is used whereas if market consistent discount rate is used, then assets are taken at market value. [1]

In an asset based discount rate approach, the investment return likely to be earned on existing assets is used as a starting point in selecting the discount rate. [0.5]

It takes into account the current asset allocation and the market yield of various asset classes to arrive at the discount rate [0.25]

- For e.g. for fixed interest rate securities it may be Gross Redemption Yield; for equities the yield implied by market price. [0.25]

The rate of discount has to be moderated (reduced) to allow for the risks associated with each asset class. [0.25]

- For example, the yield on corporate bonds has to allow for risk of default and needs to consider rating of such bonds; the market yield of equities needs to allow for the volatilities of returns. [0.5]

The gap between the market return & the discount rate will also reflect the Trustees' view on risks and future investment scenario of the scheme & the likely investment performance of each asset class & the future investment strategy likely to be adopted by them. [0.5]

- For example, a widening gap indicates likely adverse investment conditions as viewed by Trustees & their future investment strategy to back the benefits with Government Bonds [0.5]

When a market Consistent approach is used in selecting discount rate, it is assumed that it is possible to find a portfolio of Government bonds that precisely match with Pension Benefits. [0.5]

Significant risk will remain if the portfolio of bonds do not match with member's pension. [0.25]

- For e.g. the assets of suitable duration not available to match the longevity of pensioners or to match the implied guarantee in the "minimum" pension. [0.25]

An allowance has to be given for the unhedged risk by (reducing) moderating the discount rate chosen. [0.25]
[5]

Reasons for the variance in the discount rate may be as under:

- The risk of default of corporate bonds may reduce discount rate from current investment return of the fund [0.5]
- The current asset allocation may be different from the likely investment portfolio of the scheme considered for deciding the discount rate [0.5]
- The future investment return scenario considered for deciding discount rate (particularly when current asset duration is different from liability duration) may be different from current returns of portfolio [0.5]
- The discount rate in case of market consistent approach may be different from 10 year G-Sec as the liability duration may be different than 10 year [0.5]
- Discount rate considered may have margin for prudence which may differ according to valuation purpose, method and basis. In case of funding purpose or strong valuation method (say, Attained Age) and/or basis, the margin of prudence in discount rate may be lower or vice versa [1]

[3]

v) Insurance Products – features, merits & demerits

[For each product mentioned 0.5, and for features, merits and demerits of each product max 1 mark]

No	Product	Features	merit	demerit
1	Group Insurance Product (Group Term Life Insurance) [0.5]	- The policy provides death cover by charging premium based on average age & past claim experience of the group - Premium is reviewed based on scheme experience. - There may be profit sharing arrangements -Death cover may either be flat or decreasing/increasing and depending on salary/future service hence cover may be provided on need basis [1]	- It provides stability of death claim experience particularly when group size is small - Cover is available to all members up to a limit without individual underwriting hence an uninsurable member may also be covered - Premium rates are quite competitive particularly for medium to large groups [1]	- The insurer charges for his expenses & profit margin. Therefore the benefit of stability comes at an additional cost. - However the competition in this segment restricts insurer from excessive charging [1]
2	Fund based Group Products [0.5]	- Insurer takes the role of Fund Management on pooled basis. - Insurers offer both the traditional and Unit-linked contracts -Interest rates are declared based on the performance of the funds in case of traditional plans. - Charges are generally transparent, explicit. [1]	- Being the professional fund manager, the insurer will be able to provide the best balance between risk & return. - Insurance products provide desirable spread particularly to small funds - The returns are generally smoothed in case of traditional plans - Further due to competition, insurers offer attractive/competitive returns [1]	- There is a cost to the company due to the charges levied. - There could be restrictions on the access to the funds invested due to the surrender penalty imposed. - However the competition in this segment restricts insurer from excessive charging [1]

3	Group Annuity Products(immediate/Deferred) (1)	- In return for a single premium, the insurer provides a regular guaranteed income to the annuitants till their life time. - In case of deferred annuities annual premium may also be paid [1]	- The longevity risk of the pensioners is passed on to the insurer. - This will improve the security of the benefits relating to active members. [1]	- While transferring longevity risk, the benefits of outperformance of the funds will also be going to insurer. If the insurer defaults in making pension payments, the company/ Trustees may still be held responsible for the default. [1]
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[9]

vi) Impact of purchase of deferred annuities on Pension Cost

The following aspects are to be examined while considering the treatment of purchase of insurance policies to meet the benefit obligations:

- Whether such insurance policies meet the definition of "Qualifying Insurance Policy" as given in Section 8 Definition" of Ind AS 19 [0.5]
- The treatment of the payment of single premium to meet the pension obligation as specified in para (Insured Benefits) 46 -49 of Ind AS 19: [0.5]

a) Purchase of Deferred annuities to meet Deferred Pensioners Liabilities:

It is given that the benefits payable under the scheme are consistent with the benefits payable under the annuity plan and therefore there is a direct relationship with pension benefits. [0.5]

It is further assumed that

- Such purchases are made in the name of the deferred pensioners [0.25]
- The insurer has assumed the legal responsibility of paying the benefits [0.25]
- The Trustees have no further obligation to pay pension benefits [0.25]

The annuity purchases are to be considered as "Defined Contribution" for the purpose of accounting (as per Para 49), if the above conditions are met [0.5]

Impact on Accounting

The Liability in respect of the deferred pensioners will be removed from the fund. [0.25]

Assets will reduce equivalent to the purchase price of the deferred pensions. [0.25]

The purchase may cause deficit due to the prudent pricing (to meet the guarantee), expense & profit loading of the insurer kept in the premium. There will therefore be a settlement event accounting and any difference in the purchase price and accounting valuation worked out will flow through the P&L expense in that period [1.5]

There will be no change in the current service cost as there will be no benefit accrual (for the deferred pensioners) [0.5]

Net interest cost will increase if asset liquidation is more than the liabilities released or vice versa. [0.5]

Therefore there will be a moderate change in pension cost [0.25]

If the profile of the deferred pensioners has provided Actuarial loss" in the past, this will be passed on to the insurer. [0.25]

Equally if fund has earned higher investment return (than the discount rate), the actuarial gain will also be passed on to the insurer.. [0.25]

b) Purchase of Deferred annuities to meet Active members' liabilities

It is assumed that by purchasing the deferred annuity, the Trustees aim to secure the Prospective pension payable from the Normal Retirement Age (in respect of members closer to retirement). [0.5]

The pension benefit secured in such purchases will be more than the accrued benefits in respect of the active members. [0.5]

It is assumed that the fund has financial strength & willingness to make such advance funding and the company will be retaining the legal liability to pay pension from the Normal retirement date in respect of Active members. [0.5]

Therefore such purchases are to be considered as qualifying insurance Policy and are to be considered as "Plan Assets". [0.5]

They are to be taken as "fair value" while valuing ie the surrender value of the deferred annuities may be considered. [0.5]

Impact on Accounting:

Assuming that the Trustees/company is willing to finance such advance funding in respect of the prospective pension, there will be funding surplus. [0.5]

There will be no change in the current service cost in respect of the active members. [0.25]

The net interest cost will decrease due to the advance funding [0.25]

Actuarial gains on the asset side will also increase due to the advance funding if actual investment performance is higher (than expected.) [0.5]

Although the deferred annuities are purchased in respect of active members, the benefit obligations in respect of the members retained by the Company/Trust. [0.5]

Therefore any actuarial gains/losses arising due to changes in the demographic profile of the active members (e.g. mortality, resignation) will remain within the fund. [0.5]

The impact depends upon how the actuarial gain/losses compare against the surrender charges imposed by the company at the time of such decrements. [0.5]

[11]

[50 Marks]
